Program 9: Creating a Simple MAVEN TO GRADLE Project

Step 1: Create a new maven project in Devops by using the command

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenToGradle -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
sumanth@sumanth-VirtualBox:~/Devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenToGradle -DarchetypeArtifactI
d=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 11 MB/s)
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/sumanth/Devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenToGradle
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/sumanth/Devops/MyMavenToGradle
[INFO] ----------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ----------------------------------------------------------------------------
[INFO] Total time:  4.519 s
[INFO] Finished at: 2026-03-25T08:55:46+05:30
[INFO] ----------------------------------------------------------------------------
sumanth@sumanth-VirtualBox:~/Devops$ ls
MyGradleApp  MyMavenGuavaApp       MyMavenSeleniumApp02  MyMavenSeleniumApp04  MyMavenToGradle
MyMavenApp   MyMavenSeleniumApp01  MyMavenSeleniumApp03  MyMavenSeleniumApp05  MyToken
```

Step 2: Edit pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/maven-v4_0_0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.example1</groupId>
<artifactId>MyMavenToGradle</artifactId>
<packaging>jar</packaging>
<version>1.0-SNAPSHOT</version>
<name>MyMavenToGradle</name>
<url>http://maven.apache.org</url>
<!--  Properties: Customize Java version or plugin versions  -->
<properties>
<maven.compiler.source>11</maven.compiler.source>
<maven.compiler.target>11</maven.compiler.target>
```

```
</properties>
<dependencies>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>3.8.1</version>
<scope>test</scope>
</dependency>
</dependencies>
<!--  Build: Configuring plugins and build settings  -->
<build>
<plugins>
<!--  Example: Maven Compiler Plugin to compile Java code  -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.6</source>
<target>1.6</target>
</configuration>
</plugin>
<!--  Example: Maven Surefire Plugin to run tests  -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-jar-plugin</artifactId>
<version>3.1.0</version>
<configuration>
<archive>
<manifestEntries>
<Main-Class>com.example1.App</Main-Class>
</manifestEntries>
</archive>
</configuration>
</plugin>
</plugins>
</build>
</project>
```

Step 3: Execute the Tree command to display the Tree

```
sumanth@sumanth-VirtualBox:~/Devops$ cd MyMavenToGradle
sumanth@sumanth-VirtualBox:~/Devops/MyMavenToGradle$ ls
pom.xml  src
sumanth@sumanth-VirtualBox:~/Devops/MyMavenToGradle$ tree
.
├── pom.xml
└── src
    ├── main
    │   └── java
    │       └── com
    │           └── example
    │               └── App.java
    └── test
        └── java
            └── com
                └── example
                    └── AppTest.java

9 directories, 3 files
```

Step 4: Execute the command mvn clean install

```
sumanth@sumanth-VirtualBox:~/Devops/MyMavenToGradle$ gedit pom.xml
sumanth@sumanth-VirtualBox:~/Devops/MyMavenToGradle$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:MyMavenToGradle >--------------------
[INFO] Building MyMavenToGradle 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenToGradle ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenToGradle ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/sumanth/Devops/MyMavenToGradle/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenToGradle ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/sumanth/Devops/MyMavenToGradle/target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenToGradle ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/sumanth/Devops/MyMavenToGradle/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenToGradle ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/sumanth/Devops/MyMavenToGradle/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenToGradle ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.02 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenToGradle ---
[INFO] Building jar: /home/sumanth/Devops/MyMavenToGradle/target/MyMavenToGradle-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenToGradle ---
[INFO] Installing /home/sumanth/Devops/MyMavenToGradle/target/MyMavenToGradle-1.0-SNAPSHOT.jar to /home/sumanth/.m2/repository/com/example/MyMavenToGradle/1.0-SNAPSHOT/MyMavenToGradle-1.0-SNAPSHOT.ja
r
[INFO] Installing /home/sumanth/Devops/MyMavenToGradle/pom.xml to /home/sumanth/.m2/repository/com/example/MyMavenToGradle/1.0-SNAPSHOT/MyMavenToGradle-1.0-SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
```

Step 5: Now execute the command gradle init –type pom, we get build success

```
sumanth@sumanth-VirtualBox:~/Devops/MyMavenToGradle$ ls
gradle  gradlew  gradlew.bat  pom.xml  src  target
sumanth@sumanth-VirtualBox:~/Devops/MyMavenToGradle$ gradle init --type pom

> Task :init
Maven to Gradle conversion is an incubating feature.


BUILD SUCCESSFUL in 3s
2 actionable tasks: 1 executed, 1 up-to-date
sumanth@sumanth-VirtualBox:~/Devops/MyMavenToGradle$ ls
build.gradle  gradle  gradlew  gradlew.bat  pom.xml  settings.gradle  src  target
```

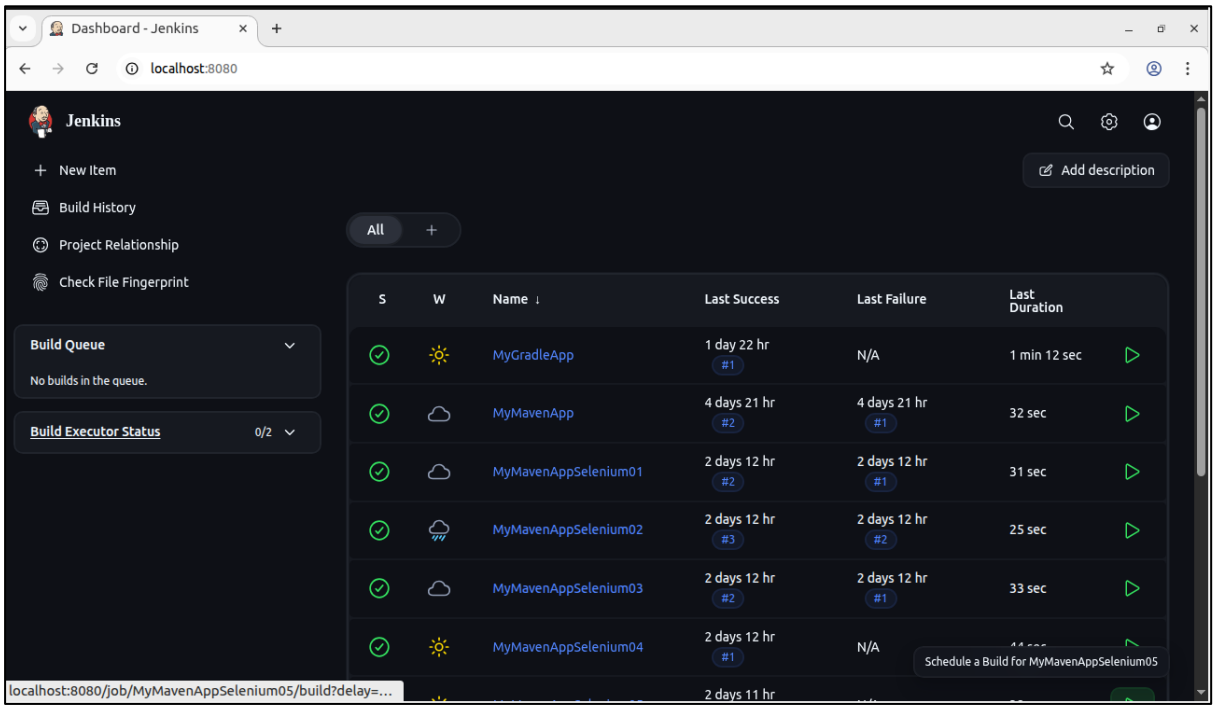
Step 6: Display the Tree Structure Generated using Tree Command

```
sumanth@sumanth-VirtualBox:~/Devops/MyMavenToGradle$ tree
.
├── build.gradle
├── gradle
│   └── wrapper
│       ├── gradle-wrapper.jar
│       └── gradle-wrapper.properties
├── gradlew
├── gradlew.bat
├── pom.xml
├── settings.gradle
├── src
│   ├── main
│   │   └── java
│   │       └── com
│   │           └── example
│   │               └── App.java
│   └── test
│       └── java
│           └── com
│               └── example
│                   └── AppTest.java
└── target
    ├── classes
    │   └── com
    │       └── example
    │           └── App.class
    ├── generated-sources
    │   └── annotations
    ├── generated-test-sources
    │   └── test-annotations
    ├── maven-archiver
    │   └── pom.properties
    ├── maven-status
    │   └── maven-compiler-plugin
    │       ├── compile
    │       │   └── default-compile
    │       │       ├── createdFiles.lst
    │       │       └── inputFiles.lst
    │       └── testCompile
    │           └── default-testCompile
    │               ├── createdFiles.lst
    │               └── inputFiles.lst
    ├── MyMavenToGradle-1.0-SNAPSHOT.jar
    ├── surefire-reports
    │   ├── com.example.AppTest.txt
    │   └── TEST-com.example.AppTest.xml
    └── test-classes
        └── com
            └── example
                └── AppTest.class

30 directories, 19 files
```

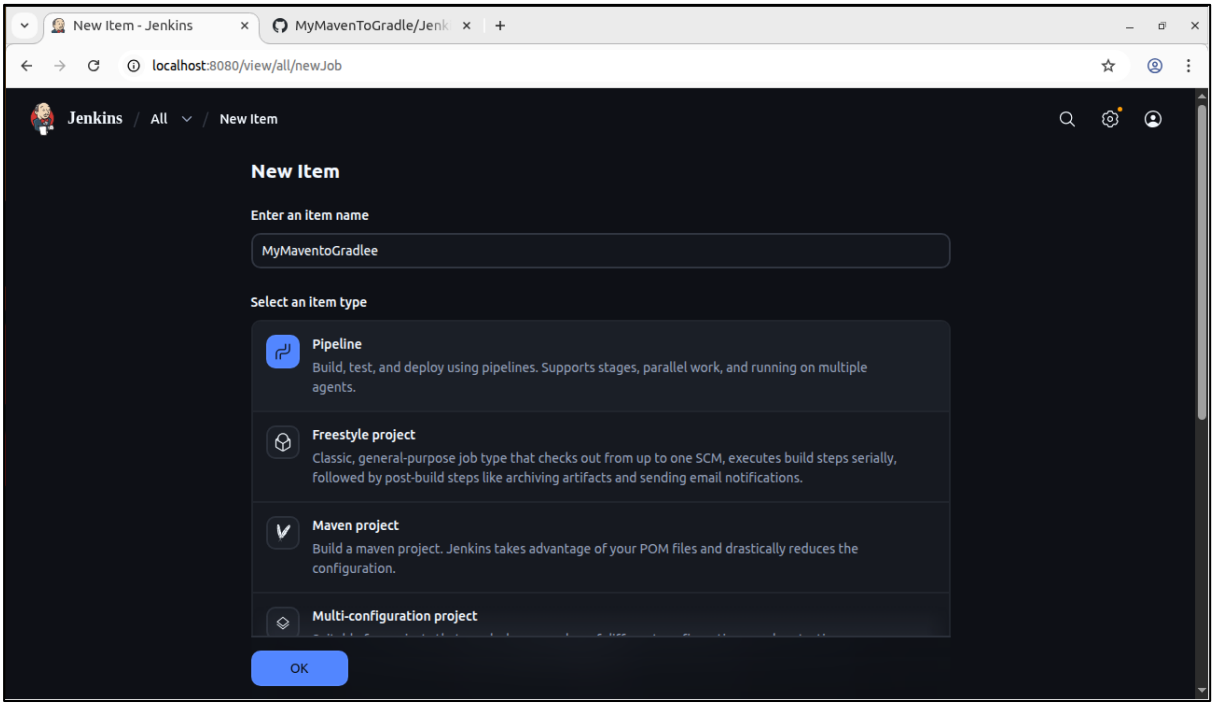
Jenkins Pipeline for MyMavenToGradle

Step 1: Open the Jenkins Dashboard in order to create a new item

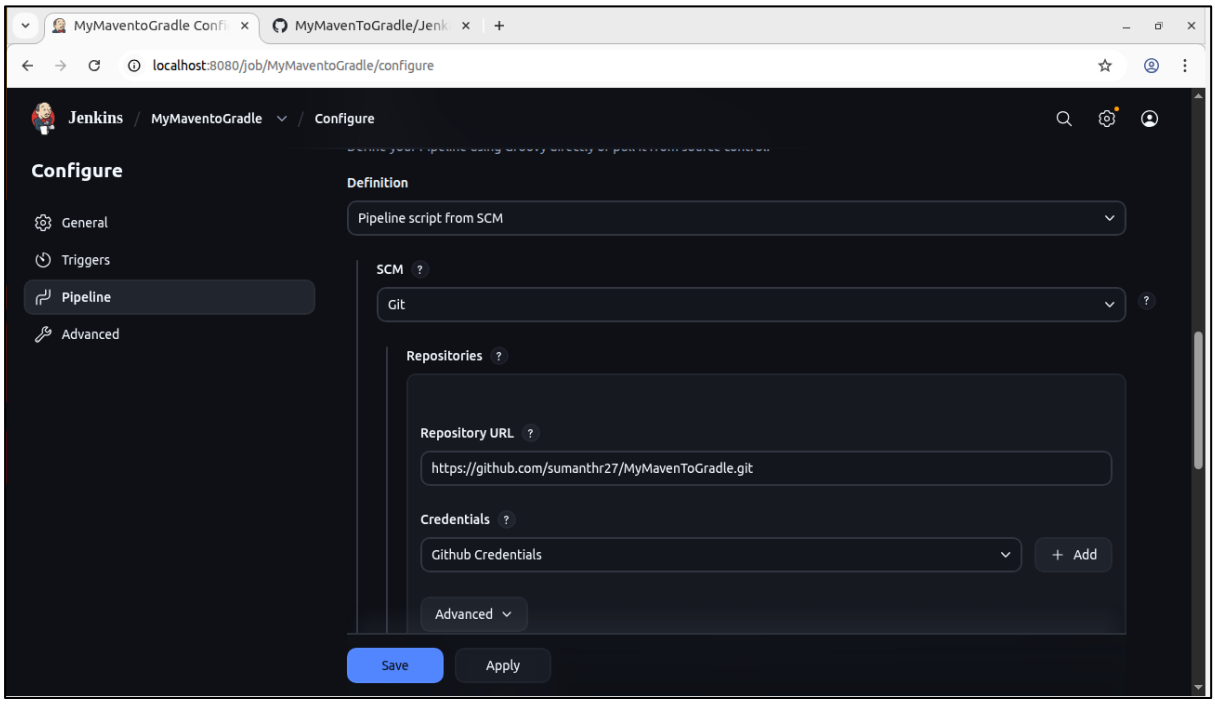


Step 2: Click on New Item in order for creating an New Item and thus give the name as “MyGradleApp” and Item type as Pipeline and

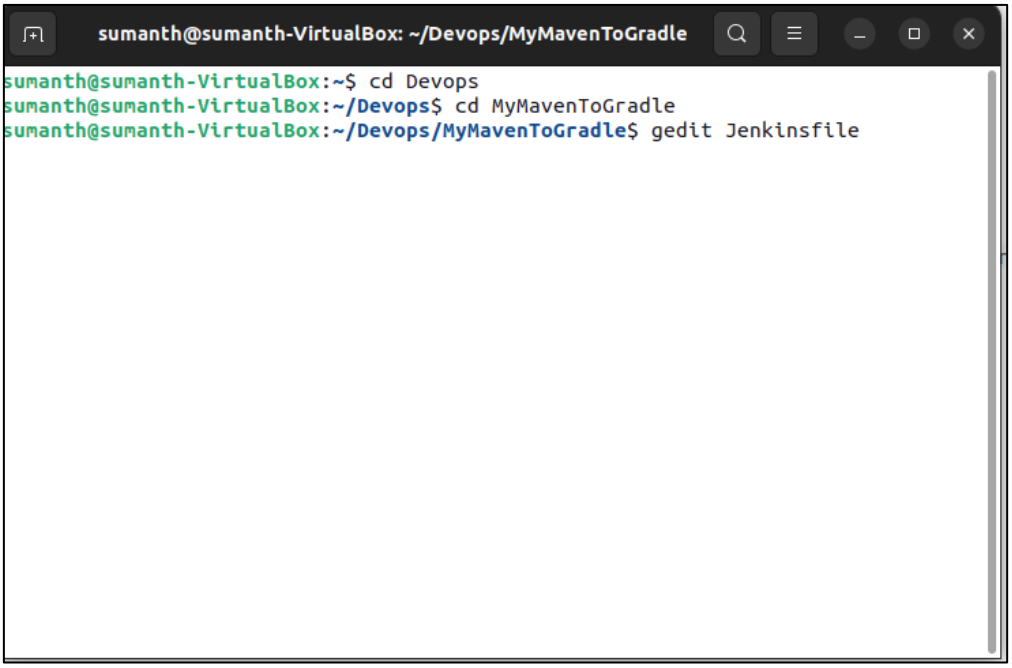
Create it



Step 3: Configure the settings by Defining the script as “Pipeline Script from SCM” and SCM as “Git”. Thus, now as we have selected Git, we need to paste the Repository URL of the project given in GitHub and choose Credentials as “GitHub Credentials” which was created earlier.



Step 4: Now in Terminal, go to directory created for MyMavenAppSelenium05 and create an new file called “Jenkinsfile”



Step 5: Paste the available code inside the Jenkinsfile

Jenkinsfile
~/Devops/MyMavenToGradle

```
pipeline {
    agent any  // Use any available agent

    tools {
        gradle 'Gradle'  // Ensure this matches the name configured in Jenkins
        jdk 'JDK'
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url: 'https://github.com/sumanthr27/MyMavenToGradle.git'
            }
        }

        stage('Build') {
            steps {
                sh 'gradle build'  // Run Gradle build
            }
        }

        stage('Test') {
            steps {
                sh 'gradle test'  // Run unit tests
            }
        }



        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'gradle run'
            }
        }
```

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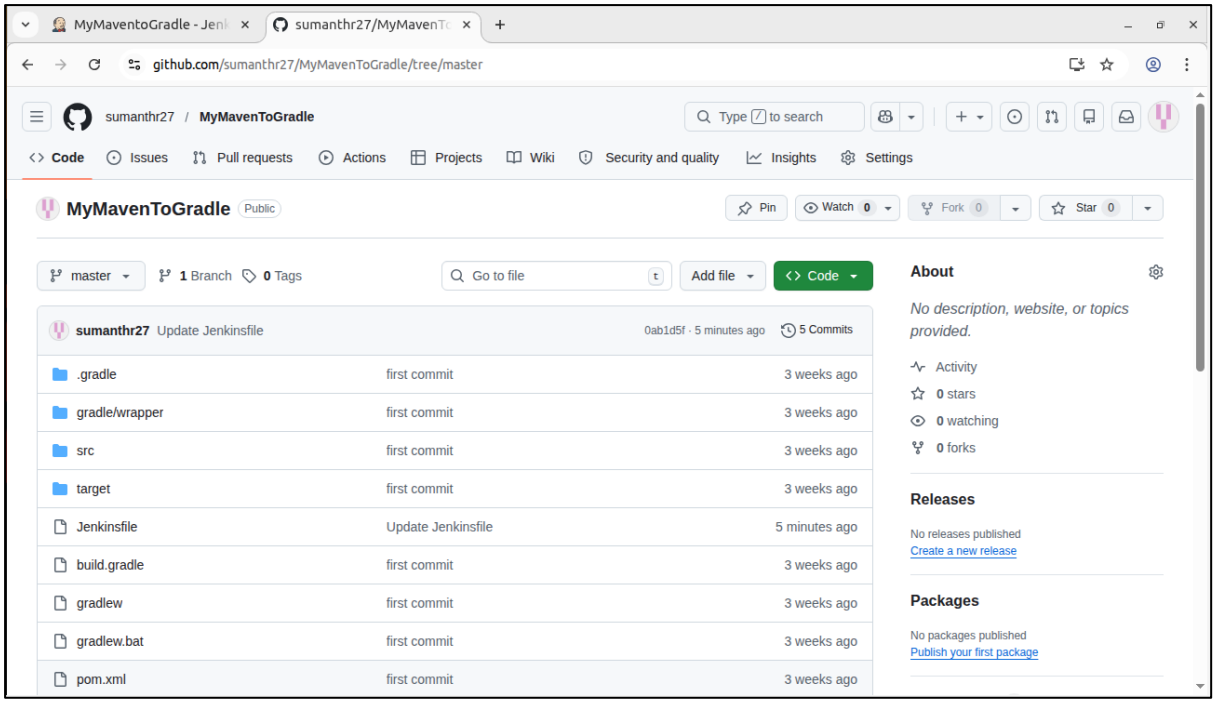
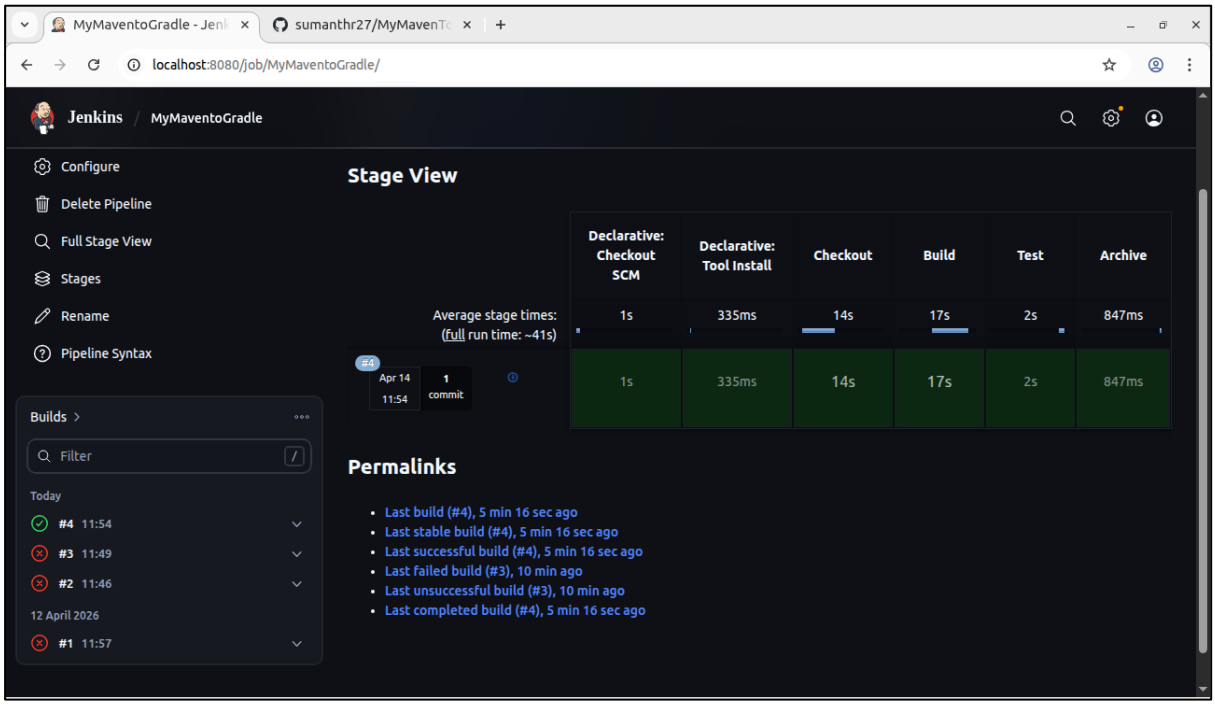
Step 6: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 7: Now we need to Push changes and commit by pushing the necessary "Jenkinsfile" to the respective Repo



Step 8: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.